

Explainable Deep Learning for Pediatric Pneumonia Detection and Classification using Chest X-ray Images

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Abstract. Pneumonia is a leading cause of mortality in children under five years globally, accounting for approximately 700,000 child deaths annually. Chest X-ray imaging remains the primary diagnostic tool; however, accurate interpretation requires skilled radiologists often unavailable in resource-limited settings. Existing deep learning systems predominantly perform binary classification without distinguishing bacterial and viral pneumonia subtypes, suffer from low accuracy on three-class tasks, and are rarely translated into clinically accessible tools. This study proposes a comprehensive deep learning framework for three-class paediatric pneumonia detection and classification (Normal, Bacterial, Viral) using the Kaggle chest X-ray dataset of 5,864 paediatric images. DenseNet121, EfficientNetV2B3, and ConvNeXtBase were experimentally trained and validated by the author on the dataset, achieving 88.3%, 87.5%, and 86.54% three-class classification accuracy respectively, outperforming prior three-class approaches. Grad-CAM explainability is integrated for transparent visual justification of predictions. A Next.js web platform and Progressive Web App mobile accessibility are currently under development by the author, with plans to provide user authentication, X-ray analysis, Grad-CAM visualisation, scan history, and patient report PDF generation. Together, these results show that accurate, explainable three-class pneumonia subtype classification is achievable, supporting the development of a practical, deployable decision-support tool for resource-limited paediatric care.

Keywords: Deep learning, Explainable AI, Paediatric pneumonia, Transfer learning, Web-based diagnosis

1 Introduction

Pneumonia remains the leading infectious cause of childhood mortality worldwide, responsible for approximately 700,000 deaths annually in children under five years of age [1]. The disease disproportionately affects low- and middle-income countries where access to trained radiologists and advanced diagnostics is severely constrained. Chest X-ray (CXR) imaging is the primary and most cost-effective diagnostic modality in clinical practice; yet its accurate interpretation demands extensive radiological expertise. In rural and resource-limited healthcare settings, the chronic scarcity of skilled radiologists frequently leads to delayed or erroneous diagnoses, contributing directly to preventable morbidity and mortality. Deep learning, and convolutional neural networks (CNNs) in particular, have demonstrated performance comparable to or exceeding experienced clinicians in a wide range of medical image classification tasks [2]. The emergence of transfer learning leveraging feature representations pre-trained on large-scale datasets such as ImageNet has made it feasible to achieve high diagnostic accuracy even when labelled medical imaging data is limited. Despite this progress, three critical

limitations persist in the literature: (i) the majority of published systems perform only binary classification, failing to distinguish bacterial from viral pneumonia, subtypes that demand fundamentally different clinical treatment strategies; (ii) the opacity of deep learning models impedes clinical adoption by undermining clinician trust in model outputs; and (iii) very few studies translate research-grade models into practically accessible clinical tools for deployment in real healthcare workflows [3].

This study addresses these limitations through an end-to-end diagnostic framework, whose primary contribution is the author's experimental training and validation of DenseNet121, EfficientNetV2B3, and ConvNeXtBase on the three-class task, achieving 88.3%, 87.5%, and 86.54% accuracy respectively. As most existing work is binary, these results advance clinically actionable pneumonia subtype differentiation. The best-performing model will be integrated into a full-featured web platform currently under development by the author, with Progressive Web App (PWA) mobile accessibility also planned to maximise clinical reach in low-resource environments.

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2 Literature review

2.1 Deep learning for pneumonia detection

Rajpurkar et al. [4] introduced CheXNet, a 121-layer DenseNet achieving radiologist-level performance on binary pneumonia detection, marking a significant milestone in the application of deep learning to chest radiograph interpretation. Subsequent transfer learning studies have demonstrated strong binary classification performance on chest X-ray datasets using various CNN architectures [5], establishing that deep learning models can reliably distinguish pneumonia-positive cases from normal chest X-rays with high accuracy. However, despite these advances, the overwhelming majority of published work frames pneumonia detection as a two-class problem, leaving a critical clinical question unaddressed. Binary classification cannot distinguish bacterial from viral pneumonia subtypes, a differentiation that is clinically essential for appropriate patient management. Bacterial pneumonia is treated with antibiotics, whereas viral pneumonia requires antiviral or supportive therapy; misclassifying one subtype as the other risks not only treatment failure but also unnecessary antibiotic use, which contributes to the broader public health challenge of antimicrobial resistance. Deep learning systems without subtype discrimination therefore offer limited clinical utility in guiding therapy decisions [6], as a positive pneumonia prediction alone is insufficient for a clinician to determine the correct course of treatment.

Published three-class pneumonia classification studies, which distinguish Normal, Bacterial, and Viral categories, remain limited in number and report substantially lower accuracy than binary approaches, reflecting the increased visual similarity between bacterial and viral presentations on chest radiographs and the inherent difficulty of the finer-grained classification task. The present work directly addresses this gap: DenseNet121 and EfficientNetV2B3 have been experimentally trained and validated by the author on the three-class task, achieving 88.3% and 87.5% accuracy respectively. As most prior work performs only binary classification, this study's three-class results represent a meaningful advance in clinically actionable pneumonia subtype differentiation.

2.2 Explainable AI and web-based clinical deployment

Selvaraju et al. [7] proposed Grad-CAM (Gradient-weighted Class Activation Mapping), which generates class-discriminative heatmaps by backpropagating gradients to the final convolutional layer and computing a weighted combination of feature maps, producing a spatial localisation map of the regions most influential for a prediction without any architectural modification to the underlying model. This technique is particularly well-suited to medical imaging, as the resulting overlays can be directly compared with radiological findings by clinicians; yet, despite its demonstrated utility, no

published study integrates Grad-CAM within a fully operational, role-based, web-deployed paediatric pneumonia classification system accessible to non-specialist clinical staff. This shortcoming reflects a broader translational gap: a systematic analysis of the deep learning medical imaging literature [8] highlights that, despite strong classification results, very few studies translate their models into deployable clinical tools. Progressive Web Apps (PWAs) offer a widely recognised approach to mobile-accessible deployment without the cost and overhead of native application development [9], yet their application to explainable medical AI systems for paediatric pneumonia diagnosis remains unexplored. Bridging this combined gap integrating explainability with accessible, web- and mobile-based clinical deployment is a core objective of the present work.

3 Methodology

3.1 Dataset and preprocessing

The Kaggle chest X-ray pneumonia dataset [10] is used, comprising 5,864 anterior-posterior CXR images of paediatric patients aged one to five years, collected from Guangzhou Women and Children's Medical Centre. Images are annotated into three categories: Normal, Bacterial Pneumonia, and Viral Pneumonia. Table 1 presents the distribution across training, validation, and test splits.

Table 1. Dataset distribution across classes.

Class	Train	Val	Test	Total
Normal	1,341	8	234	1,583
Bacterial	2,530	8	242	2,780
Viral	1,345	8	148	1,501
Total	5,216	24	624	5,864

The dataset exhibits significant class imbalance, with bacterial pneumonia comprising nearly twice the samples of normal cases. This is mitigated through data augmentation and class-weighted loss assignment. Augmentation operations applied during training include random horizontal flipping, rotation of ± 15 degrees, zoom of 10%, and brightness adjustment. All images are resized to 224×224 pixels and normalised using each model's corresponding ImageNet preprocessing function (`preprocess_input`) prior to model input, ensuring each architecture receives inputs in the distribution under which it was pre-trained. Such imbalance, if left uncorrected, would bias the model towards the majority bacterial class and inflate overall accuracy while masking poor sensitivity to the under-represented viral subtype precisely the category of greatest clinical concern. Accordingly, the class weights are assigned in inverse proportion to class frequency, and performance is reported using macro-averaged precision, recall, and F1-score, which weight every class equally so that the minority normal and viral classes contribute fully to the

evaluation rather than being overshadowed by accuracy alone.

3.2 Comparative model training

Three CNN architectures pre-trained on ImageNet are evaluated under strictly controlled, identical experimental conditions: DenseNet121, EfficientNetV2B3, and ConvNeXtBase. Each backbone is fitted with an identical regularised head (GAP $\rightarrow$ BatchNorm $\rightarrow$ Dense-512 $\rightarrow$ Dense-256 $\rightarrow$ Softmax) and trained in two phases frozen-base head training (lr 1e-3, 15 epochs), then fine-tuning of the deeper layers (lr 1e-5, 25 epochs) using class-weighted cross-entropy, early stopping, and a batch size of 16. Evaluation metrics include accuracy, precision, recall, F1-score per class, AUC-ROC, and inference time. All three architectures were fully trained and evaluated by the author, achieving 88.3% (DenseNet121), 87.5% (EfficientNetV2B3), and 86.54% (ConvNeXtBase) test accuracy on the three-class task.

3.3 Explainability with Grad-CAM

Grad-CAM [7] is the explainability method integrated into this system. For each prediction, Grad-CAM computes the gradient of the predicted class score with respect to the final convolutional layer's feature maps, then produces a weighted spatial activation map identifying which image regions most influenced the model's decision. These heatmaps are rendered as colour-coded overlays on the original CXR image and presented to the clinician alongside the classification result. The technique is architecture-agnostic, computationally efficient, and requires no modification to the trained model, making it immediately applicable to all evaluated architectures without retraining.

3.4 Web platform and mobile deployment

The diagnostic system is being developed as a Next.js web application with a Flask-based deep learning inference backend, where the frontend handles user interaction and the backend manages model inference and Grad-CAM computation. The web platform is currently under development by the author and is planned to incorporate: (1) secure user authentication with role-based access for patients, doctors, researchers, and administrators, ensuring that each user category interacts only with functionality relevant to their clinical or administrative role; (2) drag-and-drop CXR upload with optional clinical notes, providing an intuitive and efficient data entry workflow; (3) AI analysis presenting predicted class, per-class confidence scores, and Grad-CAM heatmap overlay, enabling clinicians to review both the classification outcome and its visual justification in a single interface; (4) scan history management with search, filter, and CSV export, supporting longitudinal patient monitoring and data retrieval; (5) automated patient report PDF generation including the original X-ray, heatmap overlay, prediction summary, confidence distribution, and clinical disclaimer, producing a structured and portable clinical

document; and (6) a comprehensive admin dashboard with platform statistics, user management, and activity logging, providing oversight and governance capabilities for system administrators. Progressive Web App mobile accessibility enabling home-screen installation on any device without a native application is also planned as a subsequent deployment phase, extending the system's reach to clinicians operating in mobile-primary or low-resource healthcare environments.

4 Results

All three models DenseNet121 (88.3%), EfficientNetV2B3 (87.5%), and ConvNeXtBase (86.54%) were experimentally trained and validated on the three-class task; as most prior work performs only binary classification, these subtype-differentiating results represent a meaningful contribution to the field. Table 2 reports the three-class test performance of the three evaluated architectures across five complementary metrics—accuracy, macro-averaged precision, recall, F1-score, and ROC-AUC—providing a comparison that extends beyond accuracy alone.

Table 2. Three-class test accuracy of experimentally trained and validated models in this study.

Model	Accuracy (%)	Precision	Recall	F1-score	ROC-AUC
DenseNet121	88.3	0.873	0.879	0.875	0.959
EfficientNetV2B3	87.5	0.864	0.861	0.862	0.945
ConvNeXtBase	86.54	0.852	0.855	0.854	0.925

Across all five metrics the three models follow a consistent ordering: DenseNet121 ranks highest, EfficientNetV2B3 second, and ConvNeXtBase third, yet the overall spread is narrow, with less than two percentage points separating the top and bottom accuracies (88.3% versus 86.54%). DenseNet121 obtained 88.3% accuracy with precision, recall, and F1-score of 0.873, 0.879, and 0.875 respectively and the highest ROC-AUC of 0.959; EfficientNetV2B3 followed closely at 87.5% accuracy (precision 0.864, recall 0.861, F1-score 0.862, ROC-AUC 0.945); and ConvNeXtBase reached 86.54% accuracy (precision 0.852, recall 0.855, F1-score 0.854, ROC-AUC 0.925). For every model, precision and recall remained closely balanced and the ROC-AUC exceeded 0.92, indicating strong and stable class separability rather than a tendency to favour the majority bacterial class or to neglect the minority normal and viral categories. The differences between models were modest across every metric, with the F1-score spanning only 0.875 to 0.854, so the three architectures are broadly comparable, DenseNet121 holding a consistent but small advantage. Notably, for each model the macro-averaged precision, recall, and F1-score lie within a narrow band, reflecting balanced performance across the majority bacterial and the minority normal and viral classes, and confirming that the transfer-learning pipeline generalises reliably across all three backbones.

Beyond the quantitative metrics, Grad-CAM was applied to the test predictions to examine the spatial basis of each model's decisions. For correctly classified bacterial cases, the activation maps concentrated on regions of focal consolidation and lobar opacification; viral cases produced more diffuse, bilateral activation consistent with interstitial infiltrates; and normal radiographs showed little salient activation within the lung fields. These qualitative observations indicate that all three models base their predictions on clinically relevant anatomical regions rather than background or annotation artefacts.

5 Discussion

All three models DenseNet121 (88. DenseNet121's lead reflects its densely connected design, in which feature reuse across depth and reduced vanishing gradients are particularly advantageous for the limited sample sizes typical of medical imaging, whereas EfficientNetV2B3's compound scaling, jointly optimising depth, width, and resolution, yields comparably strong features. The narrow spread between models suggests the strong performance stems mainly from the shared transfer-learning, two-phase fine-tuning, and class-weighting methodology rather than any single backbone. Because three-class classification is inherently harder than binary detection, these results represent a meaningful contribution to subtype differentiation; crucially, distinguishing bacterial from viral pneumonia supports antimicrobial stewardship in a way binary detectors cannot. As the strongest model (DenseNet121, ~8M parameters) is also the lightest relative to EfficientNetV2B3 (14M) and ConvNeXtBase (88M), it offers the most favourable accuracy-to-cost ratio for resource-constrained, mobile deployment.

Grad-CAM confirms that predictions localise clinically meaningful regions—consolidation for bacterial cases and bilateral interstitial infiltrates for viral cases—rather than dataset artefacts, though this evidence is qualitative and motivates planned quantitative (Intersection-over-Union) validation. These explainable predictions are being integrated into a secure, role-based web platform with planned mobile support; however, as all images originate from a single paediatric centre, the results constitute internal validation only and require external, multicentre confirmation before clinical deployment.

6 Conclusion and recommendation

This paper presented an explainable, three-class deep learning framework for paediatric pneumonia classification (Normal, Bacterial, Viral) that addresses five gaps in a predominantly binary literature: the absence of bacterial–viral subtype differentiation, the scarcity of validated three-class systems, the lack of controlled multi-model evaluation, the absence of integrated explainability, and the unavailability of a deployable clinical tool. Under identical experimental conditions, DenseNet121, EfficientNetV2B3, and ConvNeXtBase achieved 88.3%, 87.5%, and 86.54% three-class test accuracy respectively, with DenseNet121 the strongest

performer. Grad-CAM integration accompanies every prediction with a clinician-interpretable heatmap, directly mitigating the black-box barrier to clinical adoption. The best-performing model is being deployed through a role-based Next.js web platform providing CXR upload, three-class prediction, heatmap visualisation, scan-history management, and automated PDF reporting with Progressive Web App support planned to extend reach to low-resource, mobile-primary settings. Future work will pursue external multicentre and adult-population validation, integration of structured clinical metadata, and prospective evaluation of real-world diagnostic impact.

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